

Ajinkya Ashok

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PROFESSIONAL SUMMARY

Lead Machine Learning Engineer with 10+ years of work architecting high-performance distributed systems and production-grade AI pipelines. Experience in scaling, optimizing and automating inference and training pipelines across GCP and edge environments. Holds dual Master's degrees in Software Engineering and Cognitive Sciences, combining robust software design principles with cutting-edge machine learning systems.

TECHNICAL SKILLS

- **Infrastructure & MLOps:** CUDA, Docker, Kubernetes, GCP, Spark, Airflow, Terraform.
- **Core Capabilities:** Real-Time Inference, Distributed Computing, Model Optimization, CI/CD.
- **AI/ML Frameworks:** Python, PyTorch, TensorFlow, Scikit learn
- **Skills:** Natural Language Processing, Neural Networks, **Keras**, **OpenCv**, Kube-flow, **Dataproc**, BigQuery, VertexAI, BigQuery, Google Cloud Storage
- **Exposure:** R, Scala, Unity3D, Neuro-symbolic AI

EDUCATION

Masters in Cognitive and Information Sciences, University of California, Merced	December 2025
Masters in Software Engineering, <i>San Jose State University</i>	December 2013
Bachelors in Engineering, Information Technology, <i>University of Pune</i>	May 2008

EXPERIENCE at University of California, Merced

May 2024- present

- Developed a scalable LLM inference and evaluation framework in Python using LangChain, OpenAI APIs, asynchronous batch processing, and custom scoring modules for spatial, symbolic, and logical reasoning tasks.
- Optimized inference workflows through prompt caching, parallel API execution, token-efficient prompt templates, structured outputs, retry handling, and automated validation to reduce latency by 10%, cost, and evaluation runtime.
- Built benchmark datasets and experiment pipelines to compare model accuracy, consistency, throughput, token usage, and inference stability across model versions and configurations.
- Identified state-tracking failures and symbolic reasoning drift, generating insights that informed model selection, prompt optimization, production reliability, and deployment decisions.

WORK EXPERIENCE

CoreLogic | Sr. Machine Learning Engineer

Feb 2022 - Dec 2024

- **Built an automated ML training and validation pipeline** using GCP (Cloud Functions, Pub/Sub, BigQuery, Airflow, Dataproc), automating data ingestion, preprocessing, and model retraining. **Reduced retraining costs and standardized training/inference workflows for scalability.**
- **Developed a RAG-based Question-Answering system** using Hugging Face and a Vector Database to support domain-specific risk analysis. **Improved contextual accuracy and reliability of automated risk-profile calculations.**
- **Implemented CI/CD automation for ML pipelines** using Jenkins, Terraform, and Cloud Run, enabling automated provisioning, testing, and deployment. **Decreased deployment time by 25% and improved release stability and reproducibility.**

ChewedPixelStudios | Sr. Machine Learning Engineer

Feb 2020 - Jan 2022

- **Designed architecture for a scalable ML systems and workflows** by evaluating cloud-native components, data pipelines, and model-serving patterns, ensuring reliability, reproducibility, and seamless deployment across environments core to MLOps best practices.
- **Developed end-to-end object detection pipelines** for AR applications using Python, IBM Watson, OpenCV, TensorFlow, and Keras, automating dataset preparation, annotation, augmentation, training, and validation to support continuous model improvement.
- **Integrated trained ML models into Unity3D AR environments** and established deployment workflows enabling real-time inference, performance monitoring, and version control aligned with ML lifecycle management and **delivered production-ready ML pipelines with automated processes**.
- Deployed a YOLO computer vision model on NVIDIA Jetson, converting PyTorch to ONNX/TensorRT and applying FP16/INT8 optimization to address high edge-inference latency.
- Reduced latency by **60%** and improved throughput from **12 FPS to 30 FPS** while maintaining detection accuracy.

Wells Fargo | Natural Language Processing Engineer | San Francisco

Jan 2018 - Jan 2020

- **Migrated a conversational AI system** from regex-based logic to scalable deep learning models using Java and Bidirectional LSTMs, enabling more robust intent understanding and **increasing user engagement by 30%**.
- **Trained and optimized an ensemble of intent and entity classification models** using DL4J, OpenNLP, Python, Scikit-learn, NLTK, TensorFlow, and word vector embeddings, **improving accuracy by 25%** and **reducing response latency by 65%** compared to benchmark baselines.
- **Implemented automated ML deployment workflows** using Kafka, Kubernetes, and Docker to streamline model delivery, monitoring, and versioning aligned with MLOps best practices, **reduced commit-to-deploy time by 60%**.
- **Established scalable, production-ready model training and inference pipelines**, integrating data processing, model evaluation, and continuous optimization to support reliable real-time conversational AI capabilities.
- **Built a Voice Authentication system using Gaussian Mixture Models (GMMs)** for speaker verification, implementing feature extraction (MFCCs), model training, and real-time inference pipelines, improving authentication accuracy and enabling secure user identity validation.

Financial Engines | Machine Learning Engineer | Sunnyvale

Dec 2013 - Nov 2017

- **Designed and productionized topic modeling pipelines** using Java, Apache UIMA, Mahout, and Stanford NLP to classify product complaints and system issues. Deployed REST-based services with monitoring hooks, **reducing resolution time by 65% and while achieving 25% cost savings over projections..**
- **Implemented real-time anomaly detection systems** using Elasticsearch, Splunk, SQL, Lucene, TF-IDF, and N-gram modeling for automated log analysis and fault detection. **Decreased system failure rates by 23%** and improved incident response through alerting and dashboard integration.
- **Built and maintained automated CI/CD pipelines** using Jenkins, Docker, and Kubernetes for ML services and data workflows, ensuring reproducible builds, version-controlled deployments, and **a 60% reduction in commit-to-production time**.
- **Contributed to scalable, fault-tolerant ETL systems** using Java, SQL, REST APIs, and Jenkins to support downstream ML and analytics workloads. Focused on data reliability, recovery mechanisms, and on-call readiness aligned with SRE principles.
- **Developed observability features across ML pipelines** including metric logging, model performance monitoring, and error tracing to support continuous delivery and real-time diagnostics in production environments.